

## Supplementary Information

Best of both worlds: Upskilling professionals in data science through online coursework and in-person codeathons

Idun, Humphries, Teichman, Wright, & Hoffman, 2026

Table S1: Learner survey responses for self-assessed comfort and confidence with data science, on a 1-5 scale. SD = standard deviation.

| Measure    | Timepoint      | Mean | Median | SD    |
|------------|----------------|------|--------|-------|
| Comfort    | Pre-course     | 2.66 | 2.00   | 0.987 |
| Comfort    | Post-course    | 3.96 | 4.00   | 0.759 |
| Comfort    | Post-codeathon | 4.26 | 4.00   | 0.764 |
| Confidence | Pre-course     | 2.30 | 2.00   | 1.090 |
| Confidence | Post-course    | 3.89 | 4.00   | 0.641 |
| Confidence | Post-codeathon | 4.19 | 4.00   | 0.681 |

Table S2: Learner survey responses for self-assessed comfort and confidence with data science, paired two-sample Wilcoxon test statistics.

| Measure    | Comparison                     | Statistic | P-value | Adj. P-value |
|------------|--------------------------------|-----------|---------|--------------|
| Comfort    | Pre-course vs. Post-course     | 198       | <0.001  | <0.001       |
| Comfort    | Post-course vs. Post-codeathon | 24.5      | 0.035   | 0.142        |
| Confidence | Pre-course vs. Post-course     | 253       | <0.001  | <0.001       |
| Confidence | Post-course vs. Post-codeathon | 18        | 0.060   | 0.240        |

Table S3: Learner survey responses for self-assessed skills improvement with data science tools. SD = standard deviation.

| Timepoint      | Mean | Median | SD    |
|----------------|------|--------|-------|
| Post-course    | 4.48 | 4.00   | 0.509 |
| Post-codeathon | 4.56 | 5.00   | 0.506 |

Table S4: Fixed Effects from Linear Mixed Model: Data Cleaning and Preprocessing. SE = standard error. DF = degrees of freedom.

| Timepoint      | Estimate | SE    | DF | T Statistic | P-value | Adj. P-value |
|----------------|----------|-------|----|-------------|---------|--------------|
| Intercept      | 0.208    | 0.026 | 72 | 8.05        | <0.001  | <0.001       |
| Post-course    | 0.133    | 0.034 | 45 | 3.88        | <0.001  | 0.003        |
| Post-codeathon | 0.178    | 0.035 | 51 | 5.13        | <0.001  | <0.001       |

Table S5: Fixed Effects from Linear Mixed Model: Reproducibility and Workflow Practices. SE = standard error. DF = degrees of freedom.

| Timepoint      | Estimate | SE    | DF | T Statistic | P-value | Adj. P-value |
|----------------|----------|-------|----|-------------|---------|--------------|
| Intercept      | 0.223    | 0.027 | 76 | 8.21        | <0.001  | <0.001       |
| Post-course    | 0.188    | 0.038 | 49 | 4.96        | <0.001  | <0.001       |
| Post-codeathon | 0.279    | 0.041 | 53 | 6.77        | <0.001  | <0.001       |

Table S6: Fixed Effects from Linear Mixed Model: Analytical Reasoning and Data Exploration. SE = standard error. DF = degrees of freedom.

| Timepoint      | Estimate | SE    | DF | T Statistic | P-value | Adj. P-value |
|----------------|----------|-------|----|-------------|---------|--------------|
| Intercept      | 0.235    | 0.026 | 73 | 8.98        | <0.001  | <0.001       |
| Post-course    | 0.182    | 0.033 | 49 | 5.57        | <0.001  | <0.001       |
| Post-codeathon | 0.246    | 0.034 | 55 | 7.14        | <0.001  | <0.001       |

## R sessionInfo()

```

1 R version 4.6.0 (2026-04-24)
2 Platform: aarch64-apple-darwin23
3 Running under: macOS Sonoma 14.4.1
4
5 Matrix products: default
6 BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
7 LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
8
9 locale:
10 [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
11
12 time zone: America/New_York
13 tzcode source: internal
14
15 attached base packages:
16 [1] stats graphics grDevices utils datasets methods base
17
18 other attached packages:
19 [1] patchwork_1.3.2 lmerTest_3.2-1 emmeans_2.0.3
20 [4] lme4_2.0-1 Matrix_1.7-5 likert_1.3.5.1
21 [7] xtable_1.8-8 googlesheets4_1.1.2 lubridate_1.9.5
22 [10] forcats_1.0.1 stringr_1.6.0 dplyr_1.2.1
23 [13] purrr_1.2.2 readr_2.2.0 tidyr_1.3.2
24 [16] tibble_3.3.1 ggplot2_4.0.3 tidyverse_2.0.0
25
26 loaded via a namespace (and not attached):
27 [1] tidyselect_1.2.1 psych_2.6.5 farver_2.1.2
28 [4] S7_0.2.2 fastmap_1.2.0 digest_0.6.39
29 [7] timechange_0.4.0 estimability_1.5.1 lifecycle_1.0.5
30 [10] magrittr_2.0.5 compiler_4.6.0 rlang_1.2.0
31 [13] tools_4.6.0 utf8_1.2.6 yaml_2.3.12
32 [16] knitr_1.51 labeling_0.4.3 askpass_1.2.1
33 [19] mnormt_2.1.2 curl_7.1.0 plyr_1.8.9
34 [22] RColorBrewer_1.1-3 pkgload_1.5.2 numDeriv_2016.8-1.1
35 [25] withr_3.0.2 grid_4.6.0 googledrive_2.1.2
36 [28] scales_1.4.0 MASS_7.3-65 cli_3.6.6
37 [31] mvtnorm_1.4-1 rmarkdown_2.31 ragg_1.5.2

```

|    |      |                                   |                                   |                                 |
|----|------|-----------------------------------|-----------------------------------|---------------------------------|
| 38 | [34] | <a href="#">reformulas_0.4.4</a>  | <a href="#">generics_0.1.4</a>    | <a href="#">otel_0.2.0</a>      |
| 39 | [37] | <a href="#">rstudioapi_0.19.0</a> | <a href="#">httr_1.4.8</a>        | <a href="#">reshape2_1.4.5</a>  |
| 40 | [40] | <a href="#">tzdb_0.5.0</a>        | <a href="#">minqa_1.2.8</a>       | <a href="#">splines_4.6.0</a>   |
| 41 | [43] | <a href="#">parallel_4.6.0</a>    | <a href="#">cellranger_1.1.0</a>  | <a href="#">vctrs_0.7.3</a>     |
| 42 | [46] | <a href="#">boot_1.3-32</a>       | <a href="#">jsonlite_2.0.0</a>    | <a href="#">hms_1.1.4</a>       |
| 43 | [49] | <a href="#">pbkrtest_0.5.5</a>    | <a href="#">systemfonts_1.3.2</a> | <a href="#">glue_1.8.1</a>      |
| 44 | [52] | <a href="#">nloptr_2.2.1</a>      | <a href="#">cowplot_1.2.0</a>     | <a href="#">stringi_1.8.7</a>   |
| 45 | [55] | <a href="#">gtable_0.3.6</a>      | <a href="#">pillar_1.11.1</a>     | <a href="#">rappdirs_0.3.4</a>  |
| 46 | [58] | <a href="#">htmltools_0.5.9</a>   | <a href="#">openssl_2.4.2</a>     | <a href="#">R6_2.6.1</a>        |
| 47 | [61] | <a href="#">textshaping_1.0.5</a> | <a href="#">Rdpack_2.6.6</a>      | <a href="#">evaluate_1.0.5</a>  |
| 48 | [64] | <a href="#">lattice_0.22-9</a>    | <a href="#">rbibutils_2.4.1</a>   | <a href="#">backports_1.5.1</a> |
| 49 | [67] | <a href="#">broom_1.0.13</a>      | <a href="#">gargle_1.6.1</a>      | <a href="#">Rcpp_1.1.1-1.1</a>  |
| 50 | [70] | <a href="#">gridExtra_2.3</a>     | <a href="#">nlme_3.1-169</a>      | <a href="#">xfun_0.58</a>       |
| 51 | [73] | <a href="#">fs_2.1.0</a>          | <a href="#">pkgconfig_2.0.3</a>   |                                 |